

SSCTAD: Semi-Supervised Contrastive Learning for Trajectory Anomaly Detection

Jiaqi Zhang^{ID}, Zipei Fan^{ID}, Xuan Song^{ID}, and Ryosuke Shibasaki

Abstract—Trajectory anomaly detection is critical for transportation safety, solving traffic congestion, optimizing routes, and other smart city applications. Existing methods often misclassify reasonable but non-standard trajectories, such as route changes due to traffic congestion, as anomalies. Additionally, many methods require extensive labeled datasets for training, however, in real-world scenarios, trajectory anomaly labeled data are often scarce and costly to obtain. To address the challenges, we investigate a Semi-Supervised Contrastive learning approach for Trajectory Anomaly Detection, namely SSCTAD. Specifically, SSCTAD integrates a transformer based method to process trajectories and utilizes the in-out flow encoder to capture the spatial-temporal patterns from the trajectories, which can help the model capture the traffic conditions. Then, by leveraging a small subset of labeled data along with extensive unlabeled data, SSCTAD utilizes a semi-supervised contrastive learning component enables the model to better distinguish normal and anomalous trajectories by maximizing the similarity of latent representation from the same class and diversifying those from different classes. This method improves the performance of trajectory anomaly detection and effectively reduces the dependence of the model on the labeled dataset. Finally, experiments show that SSCTAD outperforms the other baseline models.

Index Terms—Trajectory anomaly detection, semi-supervised learning, spatio-temporal patterns, contrastive learning.

I. INTRODUCTION

WITH the widespread adoption of GPS technology, massive-scale trajectory data, i.e., vehicle trajectories, are being generated at an unprecedented velocity and volume. These data contain rich behavioral signatures of human mobility and traffic dynamics, through which many Intelligent

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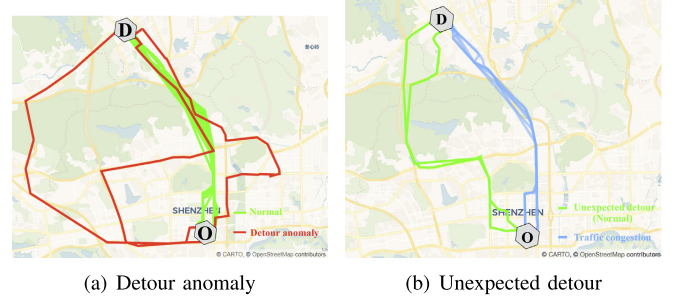


Fig. 1. The example of normal trajectories, detour anomaly trajectories and unexpected detour trajectories.

Transportation Systems (ITS) applications can be realized, such as traffic management, urban planning, public safety, and understanding urban human mobility patterns [1], [2], [3], [4].

In these various ITS applications, anomaly trajectory detection has become a critical concern to be solved in many urban scenarios [5], [6], [7], [8], [9]. This is because trajectory anomaly detection is valuable for Traffic Management, Urban Event Management, and Smart Security Center. For instance, in the Smart Security Center, the monitor can check the warning detour trajectory after a system alarm to determine whether it is necessary to take further action. Thus, the anomaly detection model can be used as a frontline filter to identify problematic trajectory behaviors, which are further processed at the Smart Safety Center.

However, detecting anomaly trajectories presents significant challenges due to the complexity and diversity of vehicle trajectories. As shown in Fig. 1, the vehicle trajectories from grid *O* to grid *D* in Shenzhen show that the vehicle trajectories are different from each other, even though they have the same starting and ending grids, which brings a significant challenge for trajectory anomaly detection.

Existing methodologies for trajectory anomaly detection often rely on predefined thresholds and distributions to identify anomalies [10], [11]. However, these methods struggle with the dynamic nature of trajectories, as trajectories are influenced by numerous unpredictable factors such as traffic congestion. Recent approaches have attempted to learn trajectory features directly from data [12], [13], [14], but they depend heavily on the quality of the labels of the datasets. This dependence makes it challenging to address complex spatio-temporal correlations, leading to potential misclassification [14], [15], [16], [17]. For instance, as shown in 1 (b), traffic congestion has occurred on the blue route, causing some vehicles to change

to a slightly farther route to avoid the congestion, which should be a normal rather than a detour anomaly.

Furthermore, GPS data usually lacks labels indicating whether a trajectory is abnormal, necessitating manual labeling by experts, which is both time-consuming and costly. Some unsupervised methods attempt to identify anomalies but are often unreliable due to assumptions about data distribution or density [11], [18], [19]. The scarcity of labels hinders the effectiveness of supervised learning approaches, which require labeled data to learn distinguishing features. To address this, researchers have proposed semi-supervised learning solutions that utilize both a small amount of labeled data and a large unlabeled dataset for training in time series classification [20], [21].

In this paper, we introduce a contrastive semi-supervised approach, SSCTAD, for trajectory anomaly detection. Unlike most existing approaches, SSCTAD utilizes a transformer-based method to learn trajectory features, which allows our model to better capture the positional information of each trajectory point. Also, we employ an in-out flow encoder to learn the traffic flow in a potential passing area, resulting in more accurate trajectory representation and reduced misclassification.

Then, to overcome the scarcity of anomaly labels, our model employs a contrastive semi-supervised learning method that enhances the ability to identify anomalous trajectories in limited labeled datasets. Specifically, we use a contrastive learning method to establish a discriminative feature space by maximizing the differences in the latent representations of trajectories across categories while minimizing the differences within each category. Through this process, our model effectively differentiates between normal, detour, and unexpected detour trajectories with limited labels. Finally, the experiments on Shenzhen and Chengdu datasets show that our proposed SSCTAD has better accuracy in trajectory anomaly detection than the baseline method.

In summary, the main contributions of SSCTAD include:

- SSCTAD utilizes a transformer based trajectory encoder with an in-out flow encoder, significantly advancing the capture of complex spatio-temporal dependencies, thereby reducing misclassification.
- SSCTAD with a contrastive semi-supervised learning approach enhancing the model's ability to discern anomalies with a limited labeled datasets. This is particularly effective in distinguishing trajectory behaviors, such as normal, detour, and unexpected detour trajectories.
- We demonstrate the efficacy of SSCTAD on Shenzhen and Chengdu trajectory datasets. The results show that SSCTAD is better than the baseline model in detecting anomaly trajectories.

The remainder of this paper is organized as follows: Section II presents a review of related work in the field of trajectory anomaly detection. In Section III, we introduce the preliminary knowledge required to understand the proposed SSCTAD framework. Section IV, we present the methodology of our proposed SSCTAD framework. Section V presents an evaluation of SSCTAD, demonstrating its better than baseline models in detecting both known and unknown anomaly

trajectories. Finally, in Section VI, we conclude the paper and discuss potential directions for future work.

II. RELATED WORK

Trajectory anomaly detection has been developed recently, driven by advances in sensing technologies and computational methods, which has made it a key aspect in the fields of traffic safety, smart city, and other directions. This section reviews the evolution of trajectory anomaly detection methods and highlights key contributions in the field.

A. Trajectory Anomaly Detection

In this section, we categorize the methods for trajectory anomaly detection into traditional approaches and deep learning approaches.

1) *Traditional Approach*: The traditional approaches to trajectory anomaly detection are based on statistical models, which use distance or density metrics to produce a deviation score between normal and anomaly trajectories. For example, Guo et al. [10] proposed a kinematic estimation-based method to detect anomalies in vessel trajectories. Lee et al. [18] introduced a hybrid method combining distance-based and density-based approaches to detect anomalous trajectories. As anomaly trajectories are few and significantly dissimilar from normal trajectories, also with the development of machine learning, researchers are utilizing machine learning to discover anomaly trajectories. Such as Witayangkurn et al. [22] proposed an anomaly detection method based on the Hidden Markov Model (HMM), aiming to detect anomaly events by identifying abnormal changes in trajectory speed and direction in urban areas. Zhang et al. [23] introduced the isolation-based method, called iBAT, to detect anomaly trajectories. However, these traditional methods can effectively detect simple anomalies but cannot identify complex anomaly trajectories. For example, the model will misclassify the unexpected detour trajectories (driving route changes due to traffic congestion or road closed) as anomalies.

2) *Deep Learning Approach*: Deep learning methods have become more prevalent in trajectory-based applications due to the fact that they can explore the rich context of the underlying trajectory with remarkable results. We will introduce these deep learning methods divided into supervised and unsupervised learning methods.

In **supervised learning**, it requires labeled data to classify trajectories as either normal or anomalous. It can efficiently capture complex patterns in the trajectory data. Such as Choi et al. [17] demonstrated that CNN is highly capable of learning representations of large-scale sequences and identifying anomalies in the data. Trinh et al. [24] employed LSTM to process the trajectory in a sequential and recursive manner to establish the anomaly detection system. Song et al. [19] utilized labeled data and constructed a supervised learning method based on RNN to detect anomaly trajectories. Both of these methods have successfully captured the complex spatio-temporal dependence of vehicle trajectories. While supervised methods often achieve high accuracy, they depend heavily on the availability of large labeled datasets, which can be

challenging to obtain in real-world scenarios, particularly for rare or novel anomalies.

In contrast to supervised learning, **unsupervised learning** methods do not require labeled data. These methods are particularly useful when labeled data is scarce or unavailable. Among them, many recent anomaly detection methods are built on generative inference networks, especially Variational AutoEncoder (VAE)-based methods. The VAE consists of a generative and inference network, which enables the model to capture complex spatio-temporal structure and to detect anomalies by identifying deviations from the learned data distribution. Several recent methods employ VAE variants to achieve anomaly trajectory detection. These methods encode trajectories into a low-dimensional latent space and identify anomalous behaviors by detecting trajectories that deviate from latent clusters. For example, Liu et al. [25] and Xie et al. [26] utilized a Gaussian Mixture Variational AutoEncoder (GM-VSAE) based method to map each sequence into latent variables that follow a mixture of Gaussian prior, which allows the model to represent multiple behavioral modes and detect anomalous as low-probability samples in the latent space. Han et al. [27] employed DeepTEA, a VAE-based method, to identify the anomaly trajectories. It learn the latent patterns from trajectories during the travel time, and uses the learned latent patterns to detect anomalies from trajectories. Additionally, Yu and Huang [28] employed a VAE-based method, where the anomaly degree is quantified based on the reconstruction probability of driving behavior feature vectors.

Above all, trajectory anomaly detection methods have evolved from traditional methods and machine learning to more advanced deep learning approaches. Although each method has strengths and achieves better results, there are still some challenges with these methods. One of the main challenges is that they tend to misclassify unexpected detour trajectories as anomaly trajectories. Additionally, many methods require extensive labeled datasets to achieve high anomaly detection accuracy, which is often scarce in real-world applications. In contrast, SSCTAD integrates a transformer based method to process trajectories and utilizes the in-out flow encoder to capture the spatial-temporal patterns from the trajectories, which can help the model capture the traffic conditions. Then, by leveraging a small subset of labeled data along with extensive unlabeled data, SSCTAD utilizes a contrastive semi-supervised learning component, which enables the model to better distinguish normal and anomalous trajectories by maximizing the similarity of latent representation from the same class and diversifying those from different classes.

B. Semi-Supervised Learning for Anomaly Detection

The challenge of limited labeled data in anomaly detection has led to the exploration of semi-supervised learning (SSL) techniques. SSL offers a promising avenue by leveraging both labeled and unlabeled data to improve model performance, making it particularly well-suited for scenarios where labeled data are scarce, costly, or difficult to obtain.

The fundamental premise of semi-supervised learning is that the vast amounts of unlabeled data contain valuable information that, when appropriately harnessed, can significantly

enhance the learning process. This is particularly relevant in anomaly detection, where the anomalies themselves are rare and diverse, making it challenging to collect sufficient labeled examples for effective supervised learning. For example, Zhu et al. [29] were among the early pioneers in SSL, introducing a graph-based approach that propagates labels from a small set of labeled samples to a large set of unlabeled data. This method demonstrated that by exploiting the structure of the data, one could achieve substantial improvements in classification accuracy, even with limited labeled data. In the specific context of trajectory anomaly detection, SSL has been employed to address the scarcity of labeled anomaly data.

In the specific context of trajectory anomaly detection, SSL has been employed to address the scarcity of labeled anomaly data. Sillito and Fisher [30] explored this approach by developing a semi-supervised learning framework tailored to the unique challenges of trajectory data in a video surveillance scenario. His work leverages a small set of labeled normal trajectories and a large pool of unlabeled data to build a model that can identify anomalies. This method effectively reduces the reliance on labeled anomaly data, allowing the model to generalize better to unseen anomalies.

In the context of trajectory anomaly detection, although SSL has emerged as a powerful tool for anomaly detection, particularly in addressing the challenge of limited labeled data. However, in trajectory anomaly detection, it may often fail to incorporate real-time contextual information, such as traffic conditions, which are crucial for accurately identifying anomalies in dynamic urban environments.

So, in our work, we propose a novel deep learning framework, SSCTAD, which employs a trajectory transformer with contrastive semi-supervised learning to address these challenges. The framework is designed to address the unique characteristics and complexities inherent in trajectory data, and the training process does not depend on extensive labeled data. Additionally, SSCTAD integrates in-out flows with the model, which addresses the limitations of both traditional anomaly detection methods and existing SSL frameworks, offering a more robust solution for dynamic urban environments.

III. PRELIMINARIES

This section provides the preliminary knowledge required to understand the proposed SSCTAD framework. First, we provide definitions related to trajectory and anomaly trajectory. Then, we formally define the problem of anomaly trajectory detection in urban areas.

A. Definitions

Definition 1 (GPS Trajectory): A GPS trajectory is a sequence of coordinates recorded by a GPS device during a trip. Formally, a trajectory, denoted as T , is a sequence of spatial-temporal points $T = (p_1, p_2, \dots, p_i)$. Each point p_i within this trajectory is represented as a tuple (lat_i, lon_i, t_i) , where lat_i and lon_i are the latitude and longitude coordinates, and t_i is the corresponding timestamp.

Definition 2 (Mapped Trajectory): In order to improve the accuracy, we use the map matching process to convert the

trajectory point p into grid cells r . For a raw GPS point p_i , the transformation can be represented by a function f_{map} , as shown in (1):

$$r_i = f_{map}(p_i) \quad (1)$$

In our paper, the grid cell information is implicitly embedded through spatial grid encoding based on the Uber H3 system. And we utilize the H3 grid cells at resolution 9, where the grid is hexagonal and has an average edge length of about 200 meters.

So, an entire trajectory $T = (p_1, p_2, \dots, p_i)$ can be transformed into $R = (r_1, r_2, \dots, r_i)$.

Definition 3 (In-Out Flow): The in-out flow of i_{th} grid r_i is characterized by the count of vehicles entering and exiting that grid over a given time slot. Formally, at the grid r_i , the in-out flow $IO(r_i)$ at time t_i is defined by the following:

$$IO(r_i) = (IO_{in}(r_i) : IO_{out}(r_i)) \quad (2)$$

Here, $IO_{in}(r_i)$ represents the number of vehicles entering grid r_i , $IO_{out}(r_i)$ represents the number of vehicles exiting grid r_i , and both are measured over the same time interval.

Definition 4 (Anomaly Trajectory): Given a trajectory T_i , which begins at region r_o and ends at region r_d , if the T_i trajectory's actual distance and the travel time are both longer than the typical or expected trajectories from the r_o to r_d during the same time period, then the trajectory T_i is a detour anomaly trajectory.

B. Problem Statements

Let c be the trajectory category labeled random variable with domain c_1, c_2, c_3 , where c_1 denotes the normal trajectories, c_2 denotes the detour anomaly trajectories, c_3 denotes the unexpected detour trajectories.

The probability of the i_{th} trajectory from area r_o to r_d is $p(T_i)$, and the conditional probability $p(T_i|c_1)$ means the probability of the trajectory T_i is a normal trajectory. Higher $p(T_i|c_1)$ indicates a higher probability that T_i is a normal trajectory. Thus, given a set of trajectories $\mathcal{T} = (T_1, T_2, \dots, T_n)$ from r_o to r_d , **our objective** is to identify whether these trajectories are normal trajectories c_1 , detour anomaly trajectories c_2 or unexpected detour trajectories c_3 .

IV. METHODOLOGY

In this section, we introduce the methodology of our proposed SSCTAD framework, as shown in Fig. 2, which contains Trajectory encoder, Inout flow encoder, Trajectory generator, Contrastive learning, and Anomaly detector.

A. Trajectory Encoder

The trajectory encoder transforms trajectory data into meaningful feature representations utilizing the convolutional neural networks and the transformer layers. This process involves several key steps:

Convolutional Embedding: In order to capture the correlation between the trajectory points and enable the model to address the irregularity of trajectory points sampling. Inspired by Traformer [31], our model utilizes the convolutional

embedding network to maintain the continuity of trajectory points across spatial and temporal domains, as:

$$\text{Conv}(r) = (f * R)(r) = \int_{-k}^k f(y)R(r_i + y)dy, \quad (3)$$

where R represents the input mapped trajectory points sequence, $f : \mathcal{F} \rightarrow \mathbb{R}$ denotes the kernel applied to the input trajectory points sequence, and $\mathcal{F} = \{-k, -k+1, \dots, k\}$. The final output of the convolutional embedding process is a set of feature vectors $\{x_1, x_2, \dots, x_n\}$.

After the convolutional embedding module, the feature vectors $\{x_1, x_2, \dots, x_n\}$ are passed to a transformer layer, effectively solving the irregularity and computational cost of trajectory data. To ensure the Transformer layer understands the order of trajectory points, the positional encoding is added to the feature vectors, which injects information about the relative position of the points in the trajectory. Here, we use sinusoidal functions, and the final input to the transformer layer is the sum of the feature vector $\mathbf{x}_i$ and its positional embedding PE_i :

$$\hat{\mathbf{x}}_i = \mathbf{x}_i + PE_i \quad (4)$$

Following positional encoding is the Transformer encoder, the core of our trajectory encoder, which leverages multi-head self-attention to capture the spatial-temporal dependencies. The self-attention mechanism computes attention scores based on three principal components derived from the input: queries (Q), keys (K), and values (V). For input vector $\hat{\mathbf{x}}_i$, such three vectors are computed by the weight matrices $\mathcal{W}_q, \mathcal{W}_k, \mathcal{W}_v$, respectively:

$$Q = \hat{\mathbf{x}}_i \mathcal{W}_q, \quad K = \hat{\mathbf{x}}_i \mathcal{W}_k, \quad V = \hat{\mathbf{x}}_i \mathcal{W}_v \quad (5)$$

Then, we perform multi-head self-attention (MSA) to capture the spatio-temporal correlations between each trajectory point. The operation is defined as:

$$\mathbf{S}^t = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (6)$$

where $\mathbf{S}^t$ is the updated features, d_k is the dimensionality of the key vectors.

After attention is computed, we employ layer normalization and a position-wise feed-forward network for each position in the sequence. Then, in order to facilitate training the transformer encoder, residual connections are employed around each of the sub-layers.

B. In-Out Flow Encoder

Inspired by the study of Zhang et al. [32], [33], our in-out flow encoder is designed to capture the dynamic flow of each urban area through the trajectories. Also, it is essential to understand the spatial and temporal dynamics of traffic flow.

Firstly, for an hourly time interval, the in-out flow can be denoted as a matrix $\mathbf{X}^h \in \mathbb{R}^{2 \times O \times D}$, where $\mathbf{X}^h$ is formed by inflow $\mathbb{I}^h \in \mathbb{R}^{O \times D}$ and outflow $\mathbb{O}^h \in \mathbb{R}^{O \times D}$. We apply convolution kernels on the in-out flow matrices to integrate the flow information and obtain their temporal embedding:

$$\hat{\mathbf{X}}^h = \text{ReLU}(\mathcal{W}_h * \mathbf{X}^h + b_h) \quad (7)$$

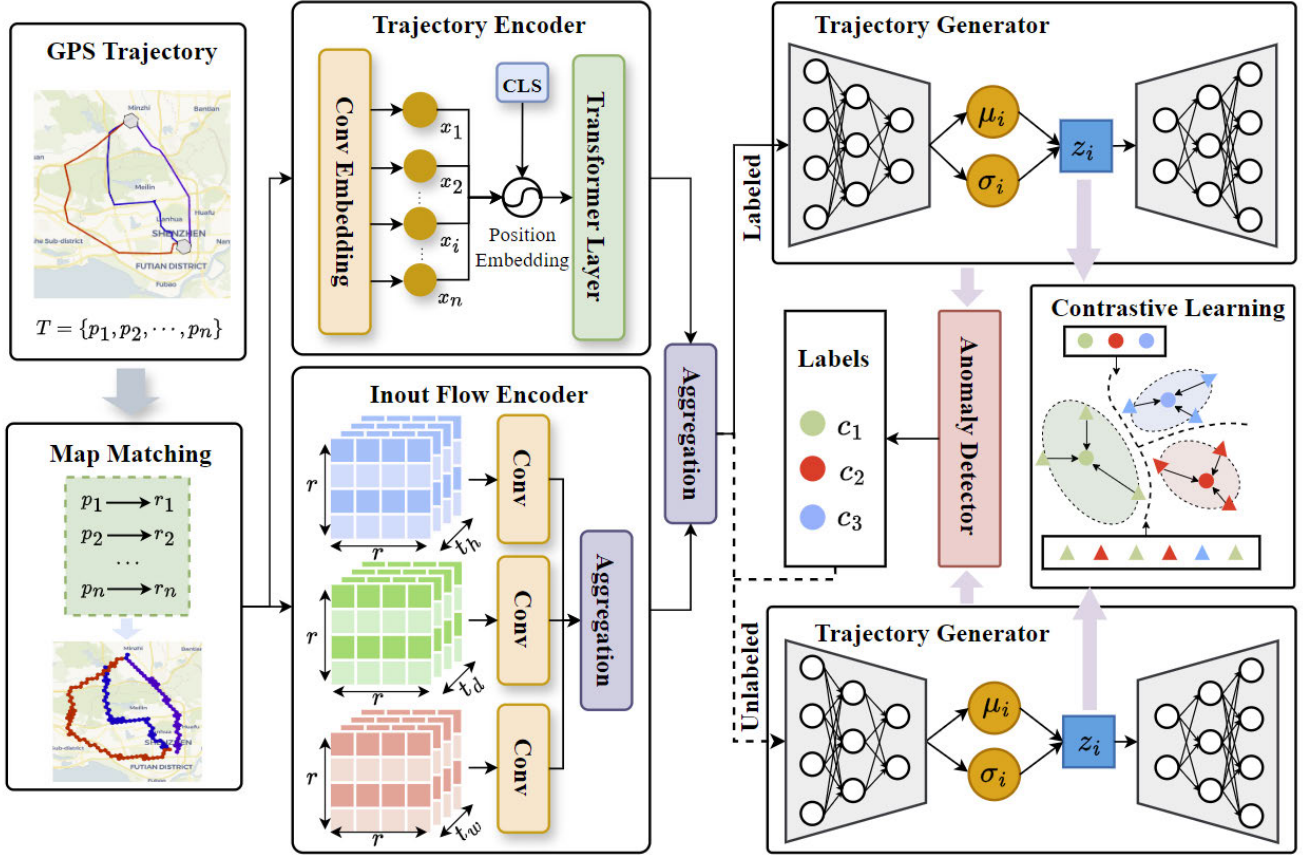


Fig. 2. The framework of SSCTAD. It consists of four main components: 1) trajectory encoder: utilize the transformer based method to process the mapped trajectory. 2) Inout flow encoder: We apply convolution kernel on the short-term and long-term in-out flow data to capture the traffic conditions. 3) Trajectory generator: based the VAE method, our method encodes each trajectory and inflow-outflow into a low-dimensional space, and enables reconstruction of the inputs by decoding. 4) Contrastive learning: our model attempts to train the trajectories with different modes to follow different Gaussian priors.

where $\mathcal{W}_h$ and b_h are learnable parameters, $*$ denotes the convolution operator, and we utilize the ReLU function as an activation function. Similarly, we can obtain the daily and weekly time interval of the in-out flow matrix $\mathbf{X}^d$ and $\mathbf{X}^w$, temporal embedding $\hat{\mathbf{X}}^d$ and $\hat{\mathbf{X}}^w$. Finally, we concatenate the above temporal embedding as follows:

$$\mathbf{S}^{io} = (\hat{\mathbf{X}}^d; \hat{\mathbf{X}}^d; \hat{\mathbf{X}}^w) \cdot \mathcal{W}_i \quad (8)$$

where $\mathcal{W}_i$ is learnable parameters, and $;$ is the concatenation operation.

C. Trajectory Generator

The trajectory generator is responsible for producing latent variable representations from the combined feature vectors $\mathbf{S}$ obtained from the trajectory encoder $\mathbf{S}^t$ and the in-out flow encoder $\mathbf{S}^{io}$. This process is crucial for capturing the underlying structure and variability in the trajectory data, which aids in anomaly detection.

Inference Net. The Inference Net in our Trajectory Generator plays a critical role in mapping the input combined feature $\mathbf{S}_i$ into a lower-dimensional latent space based on the Variational Autoencoder (VAE). This process allows us to model the distribution of latent variables z_i , which represent

the underlying structure of the trajectory T_i , and enables trajectory anomaly detection. In the context of our model, we utilize a Gaussian distribution parameterized by a mean μ_i and a standard deviation σ_i , as shown in (9).

$$z_i \sim q(z_i | \mathbf{S}_i) = \mathcal{N}(\mu_i, \sigma_i^2) \quad (9)$$

where $\mathcal{N}(\mu_i, \sigma_i^2)$ represents a Gaussian distribution with mean μ_i and standard deviation σ_i . Formally, it can be calculated as shown below.

$$\mu_i = g_1(\mathbf{S}_i) \quad (10)$$

$$\sigma_i = g_2(\mathbf{S}_i) \quad (11)$$

where $g_*(\cdot)$ is the multi-layer perceptron (MLP) function. Then, we sample from this posterior distribution using the reparameterization trick, which enables efficient gradient back-propagation during training:

$$z_i = \mu_i + \sigma_i \cdot \epsilon \quad (12)$$

where ϵ is a noise vector sampled from a standard normal distribution, formally $\epsilon \sim \mathcal{N}(0, 1)$, which allows us to generate latent variables z_i from the learned distribution.

Generative Net. Following the recent research on Variational autoencoder (VAE) [34], [35], our generative net is to

TABLE I
STATISTICS OF THE DATASETS

City	Taxi	Points	Trips
Shenzhen	20898	1197805230	6857160
Chengdu	40771	1161945300	6438300

generate the original input features through maps of the latent variables z , ensuring that the latent representation captures the essential information of the trajectory.

$$\hat{\mathbf{S}}_i = \text{ReLU}(\mathcal{W}_g z_i + b_g) \quad (13)$$

where $\mathcal{W}_g$ and b_g are learned weight matrices and bias vectors of the generative net.

Reconstruction Loss: The reconstruction loss measures the difference between the input feature vectors $\mathbf{S}$ and the reconstructed feature vectors $\hat{\mathbf{S}}$, which guides the learning of the latent representations.

$$\mathcal{L}_{\text{rec}} = \frac{1}{n} \sum_{i=1}^n \|\mathbf{s}_i - \hat{\mathbf{s}}_i\|_2^2 \quad (14)$$

Then, a regularization term based on the KL divergence is added to the loss function to ensure the latent variables follow a standard normal distribution. This term penalizes deviations from the standard normal distribution.

$$\mathcal{L}_{\text{KL}} = \text{dis}_{\text{KL}}[q_\phi(z_i|\mathbf{s}_i)||p_\theta(z_i)] \quad (15)$$

The total loss function of the trajectory generator combines the reconstruction loss and the KL divergence as follows:

$$\mathcal{L}_{\text{tg}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}} \quad (16)$$

The final output of the trajectory generator is the set of latent variables $\{z_1, z_2, \dots, z_n\}$. These latent variables provide a compact and informative representation for our trajectory anomaly detection.

D. Contrastive Learning

This subsection discusses how contrastive learning enhances the discriminative power of trajectory anomaly detection. It is designed to distinguish between normal, detour anomaly, and unexpected detour, which can effectively solve the problem of the scarcity of anomaly labels in trajectories, and the trajectory was incorrectly identified as a detour anomaly due to traffic congestion or road closure.

In contrastive net, we implicitly associate each given trajectory latent representation z_i with a pattern category c_n (e.g., Normal, Detour, or unexpected detour) and assume each pattern category to be a Gaussian distribution, as shown in Eq. (17).

$$c_n \sim \mathcal{N}(\mu_i, \sigma_i^2) \quad (17)$$

Here, μ_i and σ_i are the mean and covariance matrix of the latent representation z_i of trajectories in category c_n , respectively. Then, in order to estimate the probability of the real pattern category of a given trajectory, our model

utilizes the contrastive loss, which is designed to minimize the distance between the latent representation of a trajectory and the Gaussian distribution representing its actual category while maximizing the distance with the Gaussian distributions representing the other categories. It can be defined as Eq. (18)

$$c_n = \sum_{i=1}^i f_I[z_i \in c_n] \cdot \text{dis}_{\text{ws}}(z_i, c_n) - \log \left(\sum_{i=1}^i \exp(-\text{dis}_{\text{ws}}(z_i, c_n)) \right) \quad (18)$$

In this equation, $f_I[z_i \in c_n]$ is an indicator function that equals 1 if the real category of the given trajectory representation z_i is in c_n , and 0 otherwise. $\text{dis}_{\text{ws}}(\cdot)$ is a distance function, inspired by [35], we devise the distance by a 2-Wasserstein distance [36], [37], which is a metric in the space of probability, to measure the distance between the latent representation and the category distribution. It is defined as the minimum cost of transforming the shape of one probability distribution into the shape of the other distribution, where the cost is assumed to be the amount of distribution weight moved times the moving distance. The 2-Wasserstein distance between the distribution z_i and the distribution c_n can be calculated as Eq. (19)

$$\text{dis}_{\text{ws}}(z_i, c_n) = \|\mu_{z_i} - \mu_{c_n}\|_2^2 + \text{tr}(\Sigma_{z_i} + \Sigma_{c_n} - 2\sqrt{\Sigma_{z_i}^{1/2}\Sigma_{c_n}\Sigma_{z_i}^{1/2}}) \quad (19)$$

where $\|\cdot\|_2$ denotes the Euclidean distance, and $\text{tr}(\cdot)$ denotes the trace of a matrix.

In addition, following the existing supervised contrastive learning [35], [38], our objective function also includes a cross-entropy loss to further facilitate the training with some manually added labels (normal, detour, unexpected detour). As shown in Eq. (20).

$$\mathcal{L}_{\text{cl}} = \sum_{i=1}^i c_i \log(\hat{c}_i) \quad (20)$$

where c_i is the real label for the trajectories, $\hat{c}_i$ is the predicted trajectory label by our model, and i is the total number of categories.

E. Training and Anomaly Recognition

Training: The training process of the proposed framework is designed to optimize the model parameters by minimizing the combined loss function. Meanwhile, we also need to maximize the distance between latent variables z_i and other categories $c_{k \neq c}$.

$$\mathcal{L}_{\text{dis}}(s, c) = \frac{1}{K-1} \sum_{k \neq c}^K [l_k - \text{dis}(z_s, c_k)] \quad (21)$$

Here, l_k is the parameter that enforces all $c_{k \neq c}$ away from c_c . Above all, the total loss function integrates the $\mathcal{L}_{\text{tg}}$, $\mathcal{L}_{\text{dis}}$, and $\mathcal{L}_{\text{cl}}$, ensuring the model learns to reconstruct the input features accurately while also discriminating between normal, detour

anomaly and unexpected detour. The combined loss function is as follows:

$$\mathcal{L}_{total} = \lambda_{tg}\mathcal{L}_{tg} + \lambda_{cl}\mathcal{L}_{cl} + \lambda_{dis}\mathcal{L}_{dis} \quad (22)$$

where λ_{tg} , λ_{cl} , and λ_{dis} are the loss weights used to measure the importance of the trajectory generator, contrastive learning module, and the margin-based separation loss. At the same time, each module is crucial for the finally trajectory anomaly detection, we adopt the balanced configuration and set $\lambda_{tg} = \lambda_{cl} = \lambda_{dis} = 1$.

And our model implemented on Intel(R) Xeon(R) Silver 4214R CPU @ 2.40GHz, and NVIDIA GeForce RTX 3080 Ti (12GB) with python 3.12 and PyTorch 2.7.0. The training phase is performed using the Adam [39] optimizer with learning rate 0.001. The batch size is set to 128, and the model will trained for 500 epochs.

Anomaly Recognition: The anomaly recognition process leverages the trained model to identify anomaly trajectories. To enable the SSCTAD to identify the anomaly, we devise a probabilistic metric rule, inspired by [40], to discriminate anomalies.

Specifically, we compared the distance between latent representation z with the trajectory categories $c_k = \{c_1, c_2, c_3\}$. The distance metric d_k can be calculated as Eq. (23)

$$d_k = -\log(P_{rec}(s_i)) + dis_{ws}(z_i, c_k) \quad (23)$$

where, $P_{rec}(s_i)$ is the reconstruction probability of the input features, $dis_{ws}(z_i, c_k)$ is the Wasserstein distance between the latent representation z_i and the trajectory categories c_k . And the trajectory is assigned to the category c_k with the minimum distance, indicating the closest match among the learned categories, as shown in Eq (24).

$$\hat{c} = \arg \min_k \{d_k\} \quad (24)$$

This ensures the trajectory is classified into the most similar category in the latent space.

V. EVALUATION

A. Datasets

We conducted the experiments using two datasets. As shown in TABLE I, the Shenzhen dataset was collected from taxis between 14 October 2019 and 14 November 2019. The Chengdu dataset is collected from DiDi Chuxing ride-hailing orders¹ between 1 November 2016 and 30 November 2016. And in our dataset, only standard (non-shared) trips are included, ride-sharing orders are not present. Therefore, all trajectories in our experiments correspond to single-passenger trips, and our anomaly detection results do not consider the additional detour patterns caused by shared trips.

Following the previous studies [25], [41] and definition, we first partitioned the urban area into Uber H3.² Additionally, we analyzed each trajectory's distance, as shown in Fig. 3.

Ground Truth: Since the GPS data lacks labels (normal, detour anomaly, or unexpected detour), and inspired by the

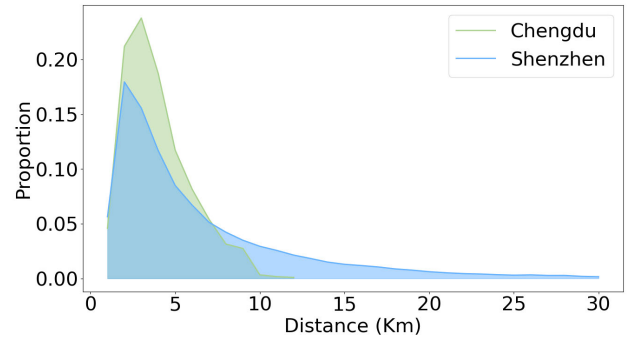


Fig. 3. The travel distance distribution of shenzhen and chengdu.

existing works [25], [27], we pick up about 10000 trajectories from the dataset and manually label whether each trajectory is normal (c_1), anomaly (c_2) or unexpected detour (c_3). The labeling process is based on a simple consensus. An example is exhibited to show how to label a trajectory. Assuming a trajectory T_i starting at r_o and ending at r_d , the traveling distance is d , and the traveling time is t_s . First, we retrieve all the trajectories from r_o to r_d , then identify the trajectory by judging the trajectory's traveling distance and time. If the trajectory's traveling distance and time are the median of the retrieved trajectories, the trajectory is **normal**. If the traveling distance and time are significantly larger than the median of the retrieved trajectories, the trajectory is **detour anomaly**. If the distance is much longer than the median, but the traveling time is closer or smaller, the trajectory is **unexpected detour**.

We split the data into a training set and a testing set, with 90% of the data used for training and 10% for testing. And we also inject 70% manually labeled data into the training set and 30% for the testing set.

B. Evaluation Metrics

To assess the performance of our model, we employed commonly used metrics from existing studies [25], [27]: Precision, Recall, F1-score, Accuracy, and ROCAUC.

- **Precision:** It uses to measure of how many of the predicted anomaly trajectories is correct (true anomaly trajectories). A higher precision indicates fewer trajectories that are falsely identified as anomalies.
- **Recall:** It uses to measure the proportion of labels correctly identified. A higher recall indicates fewer false negatives.
- **F1-score:** It is the harmonic mean of precision and recall, providing a balance between the two.
- **Accuracy:** It represents the ratio of correctly predicted instances to the total number of instances in the dataset.
- **ROCAUC** The ROCAUC (Receiver Operating Characteristic/Area Under the Curve) plot visualizes the trade-off between classifier sensitivity and specificity.

C. Baselines

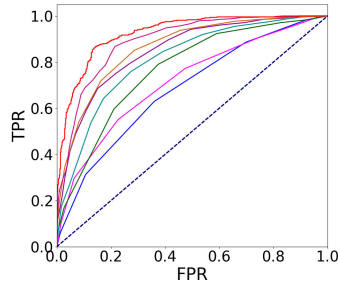
In order to demonstrate the effectiveness of our proposed model, we compare it with several existing methods for trajectory anomaly detection.

¹<https://outreach.didichuxing.com>

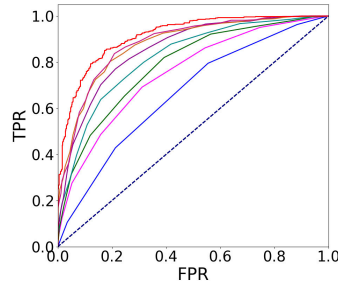
²<https://h3geo.org/>

TABLE II
PERFORMANCE COMPARISON ON SHENZHEN AND CHENGDU DATASET

Dataset	Method	Detour				Unexpected detour				Normal			
		Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
Shenzhen	TRAOD	0.7071	0.6885	0.6977	0.9135	0.6099	0.7320	0.6654	0.8053	0.8369	0.7754	0.8049	0.7604
	IBOAT	0.7119	0.7109	0.7114	0.9175	0.6265	0.7958	0.7011	0.8205	0.8592	0.7685	0.8113	0.7726
	STR	0.7303	0.6779	0.7031	0.9206	0.6689	0.7213	0.6941	0.8346	0.8601	0.8453	0.8526	0.8112
	ATD-RNN	0.7532	0.6889	0.7196	0.9261	0.6805	0.8082	0.7389	0.8507	0.8774	0.8272	0.8516	0.8154
	GM-VSAE	0.8078	0.7701	<u>0.7885</u>	<u>0.9447</u>	<u>0.7347</u>	0.7646	0.7494	0.8695	0.8955	<u>0.8892</u>	<u>0.8924</u>	0.8615
	DeepTEA	<u>0.8087</u>	0.7172	0.7602	0.9397	0.7217	<u>0.8283</u>	<u>0.7713</u>	0.8741	<u>0.9088</u>	0.8757	0.8919	0.8629
	ATROM	0.8007	<u>0.7756</u>	0.7881	0.9443	0.7371	0.8064	0.7702	<u>0.8768</u>	0.9061	0.8782	0.8919	<u>0.8631</u>
	SSCTAD	0.8371	0.8538	0.8454	0.9598	0.8149	0.8489	0.8316	0.9129	0.9369	0.9181	0.9274	0.9076
Chengdu	TRAOD	0.6915	0.6815	0.6865	0.9360	0.6591	0.6291	0.6437	0.8629	0.8739	0.8858	0.8799	0.8241
	IBOAT	0.6985	0.7065	0.7025	0.9382	0.6485	0.6389	0.6436	0.8608	0.8756	0.8778	0.8767	0.8208
	STR	0.7006	0.7126	0.7066	0.9401	0.6769	0.6438	0.6599	0.8709	0.8924	0.9018	0.8971	0.8483
	ATD-RNN	0.7561	0.7331	0.744	0.9497	0.7169	0.6947	0.7056	0.8875	0.9005	0.9111	0.9058	0.8618
	GM-VSAE	<u>0.8079</u>	0.7701	<u>0.7885</u>	0.9447	0.7348	0.7646	0.7493	0.8695	0.8955	0.8892	0.8924	0.8614
	DeepTEA	0.8033	0.7452	0.7731	0.9578	0.7838	0.7725	0.7781	0.9155	0.9295	<u>0.9413</u>	<u>0.9354</u>	<u>0.9046</u>
	ATROM	0.8041	<u>0.7704</u>	0.7869	<u>0.9597</u>	<u>0.7937</u>	<u>0.7889</u>	<u>0.7912</u>	<u>0.9201</u>	<u>0.9308</u>	0.9371	0.9339	0.9032
	SSCTAD	0.8457	0.8026	0.8236	0.9673	0.8353	0.8064	0.8206	0.9328	0.9421	0.9565	0.9493	0.9252



(a) Shenzhen.



(b) Chengdu.

Fig. 4. The ROC curve in both shenzhen and chengdu dataset.

- **TRAOD** [18]: It employs a hybrid approach, combining distance based and density based methods, within a partition and detect framework to identify trajectory outliers by considering both spatial and temporal properties.
- **IBOAT** [13]: It is a trajectory outlier detection method that utilizes an isolation based method for detection of trajectory outliers in real time.
- **STR** [12]: It is a RNN based model that leverages spatial-temporal correlations for anomaly detection in trajectory data.
- **ATD-RNN** [19]: It characterizes trajectories by learning trajectory embeddings and then employs RNN based model to capture temporal dependencies in trajectory data for anomaly detection.
- **GM-VSAE** [25]: It is a VAE based model that employs a Gaussian Mixture method to capture the multimodal distribution of trajectory data for anomaly detection.
- **DeepTEA** [27]: It is a deep learning based method that utilizes an attention mechanism with trajectory embedding to capture spatial-temporal features for anomaly detection, with a focus on critical trajectory points.
- **ATROM** [42]: It based on spatio-temporal graph convolutional adversarial network, which utilizes a

spatial-temporal graph to handle complex spatial-temporal dependencies and devises a spatio-temporal discriminator to determine whether a trajectory is anomaly or not.

D. Performance

We evaluated the performance of our model, SSCTAD, and compared it with several baseline models on two large-scale real-world datasets: Shenzhen Taxi dataset and the Chengdu DiDi dataset. The results are summarized in TABLE II, highlight the best performance in **bold** and the second-best in underlined.

In both Shenzhen and Chengdu datasets, our proposed method achieves better results for detour anomaly detection. This reflects the model's ability to identify route deviations caused by the driver's behavior accurately. Compared to the other methods, our method demonstrates a clear performance margin.

For Unexpected Detour trajectories, SSCTAD also outperforms baselines. This type of trajectory is more challenging to identify due to the fact that the route differs from the expected route, but it has to be driven an extra distance to avoid traffic

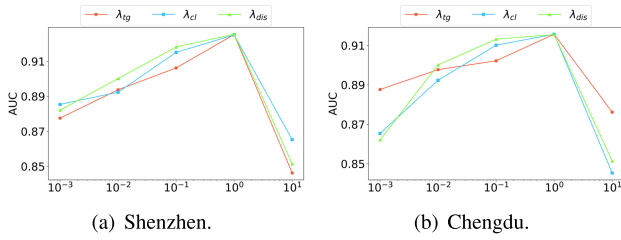
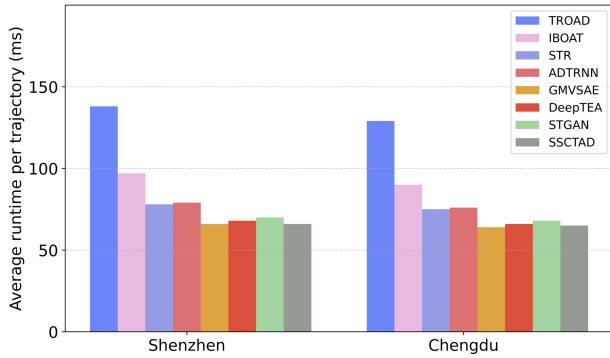
Fig. 5. The AUC performance of loss weights λ .

Fig. 6. Detection time cost comparison.

congestion. Existing methods are struggling in this category, which prefer to normalize such trajectories as anomalies. The superiority of SSCTAD highlights the effectiveness of integrating the in-out flow encoder into the learned latent space, enabling better anomaly detection results. And for identifying the normal trajectories, SSCTAD again delivers the best overall performance.

These performance improvements are attributed to the model's combination of transformer-based trajectory encoder, in-out flow encoder, and the semi-supervised contrastive learning. And by utilizing a small number of manually labeled samples, our method achieve the better trajectory anomaly detection results.

To further evaluate the discriminative capability of our proposed method, we plotted the ROC curve and calculated the Area Under the ROC Curve (AUC) for both the Shenzhen and Chengdu datasets. As shown in Fig. 4, SSCTAD achieves a higher AUC score than all baseline methods, which confirms the superior performance of SSCTAD in identifying anomaly trajectories.

Then, we also analyze the effect of loss weighting parameters during training process. As shown in Fig. 5, with the weights increase, the performance improves, and the excessively large weights lead to a decline. So, we adopt the balanced configuration and set $\lambda_{lg} = \lambda_{cl} = \lambda_{dis} = 1$.

To further evaluate the detection efficiency, we compared the average detection time per trajectory with baseline methods. For a fair comparison, all trained models were evaluated under identical conditions. The detection time cost comparison as shown in Fig. 6. Overall, the TROAD and iBOAT exhibit the longest detection times, primarily because they extract normal trajectories and compare target trajectories against these reference routes, which results in higher computational overhead.

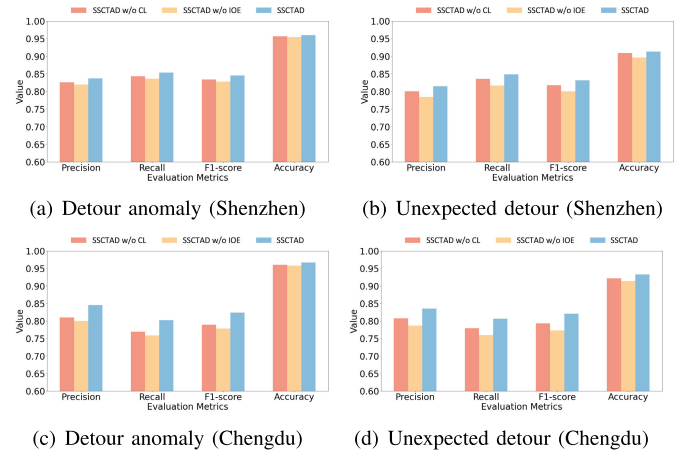


Fig. 7. The ablation study results on shenzhen and chengdu dataset.

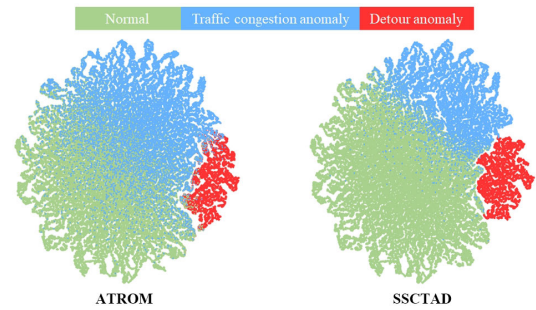


Fig. 8. Visualization of trajectory latent representation in shenzhen.

Our proposed method maintains impressive efficiency, making it well-suited for online anomaly trajectory detection scenarios.

Ablation Study: To better understand the contribution of various components in our model, we designed two modified versions: SSCTAD without the Contrastive Net (w/o CL) and SSCTAD without the in-out flow encoder (w/o IOE). As shown in Fig. 7, SSCTAD w/o IOE performs poorly in identifying unexpected detour trajectories, indicating that considering inflows and outflows in each area is crucial for detecting such anomalies. The performance of SSCTAD w/o CL is better than SSCTAD w/o IOE, but not as good as the full SSCTAD model. Removing each component leads to a performance degradation, suggesting that both the Contrastive Net and the in-out flow encoder positively impact model performance.

In the ablation study, we observed that the in-out flow encoder is particularly critical for identifying unexpected detour trajectories. This is because the encoder captures the dynamic nature of traffic flow, allowing the model to recognize deviations from typical patterns. And the Contrastive Net significantly enhances the model's ability to differentiate between normal and anomalous trajectories by learning more distinct feature representations.

Interpretability Analysis: To analyze whether SSCTAD can learn intuitive observations from an interpretable perspective, we visualized the distribution of the trajectory latent representation using the t-SNE toolkit, comparing our model and ATROM in Shenzhen over one day, as shown in



Fig. 9. Case study of shenzhen and chengdu. The green line represent normal trajectories. The red line represent detour anomaly trajectories. The blue line represent unexpected detour trajectories.

Fig. 8. The results provide significant evidence that our model effectively detects anomaly trajectories, including both detour anomalies and unexpected detour trajectories.

Case Study

Detection of Anomalous Trajectories: In this case study, we demonstrate the detection of anomalous trajectories across various travel distances using our model in both Shenzhen and Chengdu. Fig. 9 subfigures (a) to (d) depict data from a single day in Shenzhen, and subfigures (e) to (h) show data from one day in Chengdu. We detected detour trajectories and unexpected detour trajectories without manually labeled datasets using the trained model. The results are visualized with red representing detour anomalies, blue for traffic congestion trajectories, and green for normal trajectories.

Distinguishing normal and traffic congestion trajectories: As shown in Fig. 9. Traditional methods often struggle to discovery normal and traffic congestion trajectories. Our model leverages the in-out flow graph to establish the spatio-temporal relations between inflow and outflow across urban regions, enabling effective detection traffic congestion trajectories.

As shown in Fig. 10, subfigure (a) presents an example of traffic congestion trajectory detected during a morning peak hour. Subfigure (b) compares the inflow to a designated red region against its historical average. The data suggests that the morning peak hour in the red area has shifted earlier, likely due to traffic congestion in adjacent areas. This observation supports the identification of trajectories as traffic congestion trajectories.

Temporal Distribution Analysis: We further investigated the temporal distribution of normal and traffic congestion

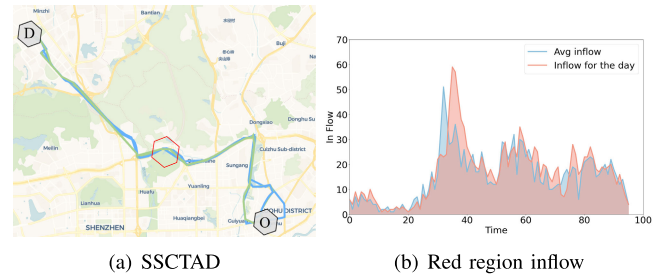


Fig. 10. An case study for our model in detecting traffic congestion trajectories.

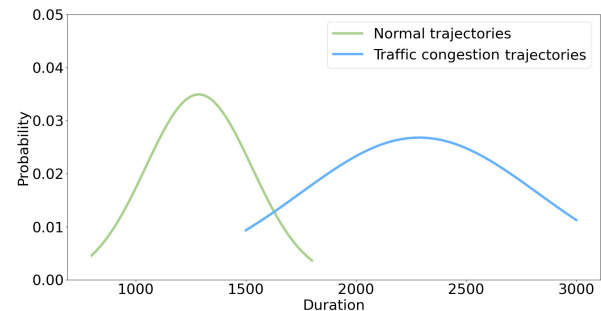


Fig. 11. Time distribution of normal trajectory and traffic congestion trajectories.

trajectories identified by our model. As shown in Fig. 11, the travel time distribution of these trajectories over a straight-line distance of approximately 12 km in Shenzhen. Detection becomes challenging for durations ranging from 1500 to

2000 seconds. Our model employs a transformer-based trajectory encoder and in-out flow encoder, combined with a contrastive learning loss based on the Wasserstein metric, to enhance anomaly detection accuracy.

In conclusion, our case study illustrates that SSCTAD effectively identifies both normal and traffic congestion trajectories. By leveraging advanced spatio-temporal modeling and visualization techniques, our model provides a robust solution for detecting and interpreting anomalous trajectories in urban environments. This capability is crucial for practical applications in urban traffic management, enabling more informed decision-making and enhancing transportation safety and efficiency.

VI. CONCLUSION

In this study, we present SSCTAD, an effective and efficient deep learning framework designed for detecting trajectory anomalies in urban environments. It utilizes trajectory encoder, In-out flow encoder, trajectory generator and trained with contrastive learning loss, thus excels in capturing the inherent complexity of spatio-temporal data, while taking into account the dynamic traffic state in urban environments. The semi-supervised contrastive learning component further enhances the model's ability to distinguish between normal trajectories, detours, and unexpected detours, especially under conditions of limited labeling data. The experimental results on large-scale real-world datasets from Shenzhen and Chengdu demonstrate that our proposed method, SSCTAD, consistently outperforms baseline model across all metrics, including precision, recall, F1-score, and accuracy. The case study further demonstrates that our model can better detect normal, traffic congestion, and detour anomaly trajectories.

In the future, we aim to enhance our model's predictive accuracy and perceptual capabilities by incorporating more features, such as special events, so that our model can detect the anomaly events. We will also incorporate an online detection module, enabling our proposed method achieve higher accuracy anomaly trajectory detection in real-time. And we also aim to extending our framework to explicitly handle shared-trip scenarios. It will present a new challenge and an interesting direction for future work, as the intentional detour and intermediate stops cause the model to incorrectly identify such trajectories as anomalies. Additionally, we plan to transfer our framework to detect the anomalies behaviors in maritime, such extensions will contribute to the development of smart ocean system construction.

REFERENCES

- [1] L. X. Pang, S. Chawla, W. Liu, and Y. Zheng, "On detection of emerging anomalous traffic patterns using GPS data," *Data Knowl. Eng.*, vol. 87, pp. 357–373, Sep. 2013.
- [2] Y. Zheng, L. Capra, O. Wolfson, and H. Yang, "Urban computing: Concepts, methodologies, and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 5, no. 3, pp. 1–55, 2014.
- [3] Q. Chen, X. Song, H. Yamada, and R. Shibasaki, "Learning deep representation from big and heterogeneous data for traffic accident inference," in *Proc. AAAI Conf. Artif. Intell.*, 2016, vol. 30, no. 1, pp. 1–7.
- [4] R. Jiang et al., "DeepCrowd: A deep model for large-scale citywide crowd density and flow prediction," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 276–290, Jan. 2023.
- [5] X. Zhang, Y. Zheng, Z. Zhao, Y. Liu, M. Blumenstein, and J. Li, "Deep learning detection of anomalous patterns from bus trajectories for traffic insight analysis," *Knowledge-Based Syst.*, vol. 217, Apr. 2021, Art. no. 106833.
- [6] M. Zhang, T. Li, Y. Yu, Y. Li, P. Hui, and Y. Zheng, "Urban anomaly analytics: Description, detection, and prediction," *IEEE Trans. Big Data*, vol. 8, no. 3, pp. 809–826, Jun. 2022.
- [7] A. Belhadi, Y. Djenouri, J. C.-W. Lin, and A. Cano, "Trajectory outlier detection: Algorithms, taxonomies, evaluation, and open challenges," *ACM Trans. Manage. Inf. Syst.*, vol. 11, no. 3, pp. 1–29, Sep. 2020.
- [8] R. Jiang et al., "DeepUrbanEvent: A system for predicting citywide crowd dynamics at big events," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery, Data Mining*, Jul. 2019, pp. 2114–2122.
- [9] Y. Zheng, "Trajectory data mining: An overview," *ACM Trans. Intell. Syst. Technol.*, vol. 6, no. 3, pp. 1–41, 2015.
- [10] S. Guo, J. Mou, L. Chen, and P. Chen, "An anomaly detection method for AIS trajectory based on kinematic interpolation," *J. Mar. Sci. Eng.*, vol. 9, no. 6, p. 609, Jun. 2021.
- [11] F. Ding, J. Wang, J. Ge, and W. Li, "Anomaly detection in large-scale trajectories using hybrid grid-based hierarchical clustering," *Int. J. Robot. Autom.*, vol. 33, no. 5, pp. 1–18, 2018.
- [12] S. Qian et al., "Detecting taxi trajectory anomaly based on spatio-temporal relations," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6883–6894, Jul. 2022.
- [13] C. Chen et al., "IBOAT: Isolation-based online anomalous trajectory detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 2, pp. 806–818, Jun. 2013.
- [14] H. Zhang, Y. Luo, Q. Yu, L. Sun, X. Li, and Z. Sun, "A framework of abnormal behavior detection and classification based on big trajectory data for mobile networks," *Secur. Commun. Netw.*, vol. 2020, pp. 1–15, Dec. 2020.
- [15] A. Vahedian, X. Zhou, L. Tong, Y. Li, and J. Luo, "Forecasting gathering events through continuous destination prediction on big trajectory data," in *Proc. 25th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2017, pp. 1–10.
- [16] M.-H. Jeong, S.-B. Jeon, S. Park, and S. Kang, "Anomaly detection in taxi flow by a projection method," *Sensors Mater.*, vol. 31, no. 11, pp. 3827–3834, 2019.
- [17] K. Choi, J. Yi, C. Park, and S. Yoon, "Deep learning for anomaly detection in time-series data: Review, analysis, and guidelines," *IEEE Access*, vol. 9, pp. 120043–120065, 2021.
- [18] J.-G. Lee, J. Han, and X. Li, "Trajectory outlier detection: A partition-and-detect framework," in *Proc. IEEE 24th Int. Conf. Data Eng.*, Apr. 2008, pp. 140–149.
- [19] L. Song, R. Wang, D. Xiao, X. Han, Y. Cai, and C. Shi, "Anomalous trajectory detection using recurrent neural network," in *Proc. Int. Conf. Adv. Data Mining Appl.* Cham, Switzerland: Springer, Nov. 2018, pp. 263–277.
- [20] H. Fan, F. Zhang, R. Wang, X. Huang, and Z. Li, "Semi-supervised time series classification by temporal relation prediction," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 3545–3549.
- [21] C. Wei, Z. Wang, J. Yuan, C. Li, and S. Chen, "Time-frequency based multi-task learning for semi-supervised time series classification," *Inf. Sci.*, vol. 619, pp. 762–780, Jan. 2023.
- [22] A. Witayangkurn, T. Horanont, Y. Sekimoto, and R. Shibasaki, "Anomalous event detection on large-scale GPS data from mobile phones using hidden Markov model and cloud platform," in *Proc. ACM Conf. Pervasive Ubiquitous Comput. Adjunct. Publ.*, 2013, pp. 1219–1228.
- [23] D. Zhang, N. Li, Z.-H. Zhou, C. Chen, L. Sun, and S. Li, "IBAT: Detecting anomalous taxi trajectories from GPS traces," in *Proc. 13th Int. Conf. Ubiquitous Comput.*, Sep. 2011, pp. 99–108.
- [24] H. D. Trinh, L. Giupponi, and P. Dini, "Urban anomaly detection by processing mobile traffic traces with LSTM neural networks," in *Proc. 16th Annu. IEEE Int. Conf. Sens., Commun., Netw. (SECON)*, Jun. 2019, pp. 1–8.
- [25] Y. Liu, K. Zhao, G. Cong, and Z. Bao, "Online anomalous trajectory detection with deep generative sequence modeling," in *Proc. IEEE 36th Int. Conf. Data Eng. (ICDE)*, Apr. 2020, pp. 949–960.
- [26] L. Xie et al., "A novel model for ship trajectory anomaly detection based on Gaussian mixture variational autoencoder," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 13826–13835, Nov. 2023.
- [27] X. Han, R. Cheng, C. Ma, and T. Grubenmann, "DeepTEA: Effective and efficient online time-dependent trajectory outlier detection," *Proc. VLDB Endowment*, vol. 15, no. 7, pp. 1493–1505, Mar. 2022.

- [28] W. Yu and Q. Huang, "A deep encoder-decoder network for anomaly detection in driving trajectory behavior under spatio-temporal context," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 115, Dec. 2022, Art. no. 103115.
- [29] X. Zhu, Z. Ghahramani, and J. D. Lafferty, "Semi-supervised learning using Gaussian fields and harmonic functions," in *Proc. 20th Int. Conf. Mach. Learn. (ICML)*, 2003, pp. 912–919.
- [30] R. R. Sillito and R. B. Fisher, "Semi-supervised learning for anomalous trajectory detection," in *Proc. Brit. Mach. Vis. Conf.*, 2008, pp. 1035–1044.
- [31] Y. Liang et al., "TrajFormer: Efficient trajectory classification with transformers," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2022, pp. 1229–1237.
- [32] J. Zhang, Y. Zheng, and D. Qi, "Deep spatio-temporal residual networks for citywide crowd flows prediction," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2017, vol. 31, no. 1, doi: [10.1609/aaai.v31i1.10735](https://doi.org/10.1609/aaai.v31i1.10735).
- [33] J. Zhang, Y. Zheng, J. Sun, and D. Qi, "Flow prediction in spatio-temporal networks based on multitask deep learning," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 3, pp. 468–478, Mar. 2020.
- [34] D. P. Kingma and M. Welling, "An introduction to variational autoencoders," *Found. Trends Mach. Learn.*, vol. 12, no. 4, pp. 307–392, Nov. 2019.
- [35] J. Bai, S. Kong, and C. P. Gomes, "Gaussian mixture variational autoencoder with contrastive learning for multi-label classification," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 1383–1398.
- [36] A. Mallasto and A. Feragen, "Learning from uncertain curves: The 2-Wasserstein metric for Gaussian processes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, J. Zhang, Y. Zheng, and D. Qi, Eds., Curran Associates, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/7a006957be65e608e863301eb98e1808-Paper.pdf
- [37] V. Masarotto, V. M. Panaretos, and Y. Zemel, "Procrustes metrics on covariance operators and optimal transportation of Gaussian processes," *Sankhya A*, vol. 81, no. 1, pp. 172–213, Feb. 2019.
- [38] P. Khosla et al., "Supervised contrastive learning," in *Proc. NIPS*, 2020, pp. 18661–18673.
- [39] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [40] J. Lu, Y. Xu, H. Li, Z. Cheng, and Y. Niu, "PMAL: Open set recognition via robust prototype mining," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 2, pp. 1872–1880.
- [41] Y. Zheng, M. Jin, Y. Liu, L. Chi, K. T. Phan, and Y. P. Chen, "Generative and contrastive self-supervised learning for graph anomaly detection," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12220–12233, Dec. 2021.
- [42] Q. Gao, X. Wang, C. Liu, G. Trajcevski, L. Huang, and F. Zhou, "Open anomalous trajectory recognition via probabilistic metric learning," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, 2023, pp. 2095–2103.

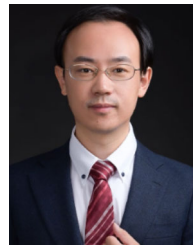


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ing, machine learning, spatio-temporal data mining, and heterogeneous data fusion.

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